

Resume Matcher

AI-powered resume tailoring engine — how the system works, end to end.

What it is

Resume Matcher is a backend API (FastAPI + LiteLLM) that stores one **master resume** per user, then uses an LLM to **tailor a copy of it to a specific job description** — rewriting bullets, injecting missing keywords, and scoring the result against an ATS-style rubric — without ever fabricating experience the candidate doesn't have. A companion **Application Tracker** follows each tailored resume through the job-search pipeline. Any frontend (web, mobile, or another backend) can drive it entirely through JSON endpoints under `/api/v1`.

Core entities

Master Resume — the one source-of-truth resume (uploaded PDF/DOCX or built via the AI Resume Wizard). Structured into sections (experience, education, projects, skills, custom sections).
Job Description — pasted JD text; AI extracts company/role/keywords once and caches them.
Tailored Resume — a per-job derivative of the master, created by the tailoring pipeline; never overwrites the master.
Application — one Kanban card per tailored resume, tracking its status through the hiring pipeline.

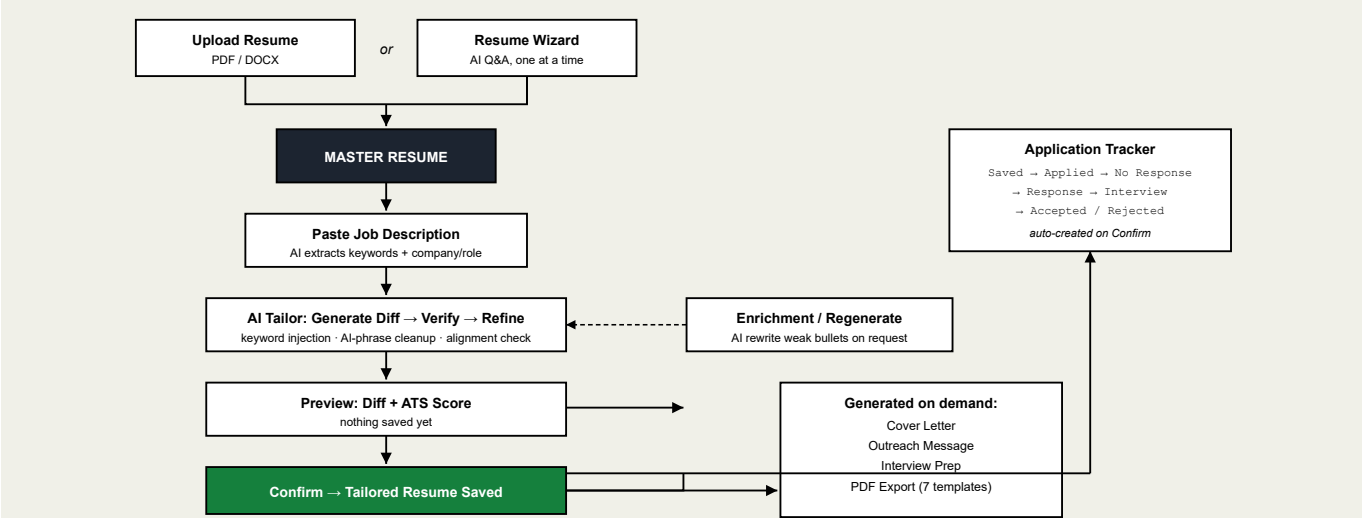
Tech stack

Backend	FastAPI, Python 3.13, LiteLLM (multi-provider AI)
Data	SQLite (async SQLAlchemy)
PDF	Headless Chromium render of print pages
Frontend	Any client consuming the JSON API (decoupled)

Safety nets (why tailoring is trustworthy)

- Diff-based edits are **verified** before being applied — personal info, dates, and skills the LLM drops are restored automatically.
- Preview → Confirm uses a content hash: confirming can only persist exactly what was previewed.
- Custom sections and unedited fields are protected from AI hallucination.

End-to-end flow



Flow stages, in short

- 1 — Get a master resume.** Upload a PDF/DOCX, or build one from scratch via the adaptive Resume Wizard.
- 2 — Paste a job description.** Keywords, company, and role are extracted once per JD and cached.
- 3 — Tailor (preview).** AI proposes targeted diffs, applies + verifies them, then refines (keyword injection, AI-phrase removal, alignment check). Nothing is persisted yet — response includes the proposed resume, a diff summary, and an ATS score.
- 4 — Confirm.** Re-validates the preview hash, then persists the tailored resume and — if enabled — generates a cover letter, outreach message, and interview prep in parallel. A tracker card is auto-created in "Applied".
- 5 — Iterate.** AI Enrichment (clarifying Q&A on weak bullets) and AI Regenerate (rewrite on instruction) refine any resume; changes are always diff-previewed before applying.
- 6 — Export & track.** Download a PDF (7 templates), and manage the application's lifecycle on the Kanban tracker until Accepted or Rejected.